

Recall expressions for gradient (2 classes ago)

$$\min_{\substack{m \\ m \in H^1}} \phi(u, m) := \frac{1}{2} \int_{\Omega} (Bu - d)^2 dx + \frac{\beta}{2} \int_{\Omega} \nabla m \cdot \nabla m dx$$

where u depends on m through the forward problem (weak form):

$$\int_{\Omega} [\underbrace{m \nabla u \cdot \nabla p}_{\text{diffusion}} + \underbrace{p \vec{v} \cdot \nabla u}_{\text{advection}} + \underbrace{p a u^3}_{\text{reaction}} - \underbrace{p f}_{\text{source}}] dx = 0 \quad \forall p \in H_0^1$$

Lagrangian:

$$\begin{aligned} \mathcal{L}(u, m, p) := & \frac{1}{2} \int_{\Omega} (Bu - d)^2 dx + \frac{\beta}{2} \int_{\Omega} \nabla m \cdot \nabla m dx \\ & + \int_{\Omega} [m \nabla u \cdot \nabla p + p \vec{v} \cdot \nabla u + p a u^3 - p f] dx \end{aligned}$$

1st Variations

1) $\delta_p \mathcal{L} = 0 \quad \forall \hat{p} \in H_0^1 \Rightarrow$ forward problem (weak form)

$$\int_{\Omega} [m \nabla u \cdot \nabla \hat{p} + \hat{p} \vec{v} \cdot \nabla u + \hat{p} a u^3 - \hat{p} f] dx = 0 \quad \forall \hat{p} \in H_0^1$$

2) $\delta_u \mathcal{L} = 0 \quad \forall \hat{u} \in H_0^1 \Rightarrow$ adjoint problem (weak form)

$$\int_{\Omega} [Bu - d] B \hat{u} + [m \nabla \hat{u} \cdot \nabla p + p \vec{v} \cdot \nabla \hat{u} + 3 p a u^2 \hat{u}] dx = 0 \quad \forall \hat{u} \in H_0^1$$

3) $\delta_m \mathcal{L} =$ weak form of gradient (Fréchet derivative of ϕ w.r.t m in direction $\hat{m}$)

$$\delta_m \mathcal{L} = \int_{\Omega} [\beta \nabla m \cdot \nabla \hat{m} + \nabla u \cdot \nabla p \hat{m}] dx = \underbrace{(\mathcal{G}(m), \hat{m})}_{\text{inner product}}$$

inner product

Derive the hessian action on $\tilde{m}$ evaluated for m , for all $\tilde{m}$:

$$\Rightarrow (\hat{m}, H(m)\tilde{m})$$

$\hat{m} \Rightarrow \tilde{m}$; weak form of gradient

$$\mathcal{L}^H(u, m, p; \tilde{u}, \tilde{m}, \tilde{p}) = \int_{\Omega} [\tilde{p} \nabla m \cdot \nabla \tilde{m} + \tilde{m} \nabla u \cdot \nabla \tilde{p}] dx + \int_{\Omega} [\tilde{m} \nabla u \cdot \nabla \tilde{p} + \tilde{p} v \cdot \nabla u + \tilde{p} a u^3 - \tilde{p} f] dx$$

start w/ gradient, add penalty of forward/adjoint problems

Lagrange multipliers for enforcing forward/adjoint equations

$\tilde{p} \Rightarrow \tilde{p}$ (Lagrange multiplier) forward problem

using $\wedge$ to indicate variations.

$$+ \int_{\Omega} [(Bu-d)B\tilde{u} + m \nabla \tilde{u} \cdot \nabla \tilde{p} + \tilde{p} v \cdot \nabla \tilde{u} + 3pa u^2 \tilde{u}] dx$$

$\tilde{u} \Rightarrow \tilde{u}$ weak form of adj.

1) $\delta_p \mathcal{L}^H = 0 \quad \forall \hat{p} \in H'_0 \Rightarrow$ incremental forward problem (tangent linear model)

find $\tilde{u} \in H'_0$
s.t.: $\delta_p \mathcal{L}^H = \int_{\Omega} [\tilde{m} \nabla u \cdot \nabla \hat{p} + m \nabla \tilde{u} \cdot \nabla \hat{p} + \hat{p} v \cdot \nabla \tilde{u} + 3\hat{p} a u^2 \tilde{u}] dx = 0$

(weak form) incremental forward problem; solve for $\tilde{u}$ (Lagrange multiplier for forward problem).

2) $\delta_u \mathcal{L}^H = 0 \quad \forall \hat{u} \in H'_0 \Rightarrow$ incremental adjoint equation/problem ("2nd order adjoint")

find $\tilde{p} \in H'_0$

s.t.: $\delta_u \mathcal{L}^H = \int_{\Omega} [\tilde{m} \nabla \hat{u} \cdot \nabla \tilde{p} + m \nabla \hat{u} \cdot \nabla \tilde{p} + \tilde{p} v \cdot \nabla \hat{u} + 3\tilde{p} a u^2 \hat{u} + (B\hat{u})B\tilde{u} + 6pa u \hat{u} \tilde{u}] dx = 0 \quad \forall \hat{u} \in H'_0$

$\times$ to make weak form

3) $\delta_m \mathcal{L}^H = (\hat{m}, H(m)\tilde{m})$ Hessian action in direction $\tilde{m}$

$$\delta_m \mathcal{L}^H = \int_{\Omega} [\tilde{p} \nabla \hat{m} \cdot \nabla \tilde{m} + \hat{m} \nabla u \cdot \nabla \tilde{p} + \tilde{m} \nabla \tilde{u} \cdot \nabla \tilde{p}] dx$$

$\forall \hat{m} \in H^1$

$$\hat{m}, H(m)\tilde{m}$$

$u, p, \tilde{u}, \tilde{p}$ depend on m
So m doesn't need to explicitly appear

Must solve: 1) forward, 2) adjoint, 3) incremental forward,
4) incremental adjoint to get $u, p, \tilde{u}, \tilde{p}$
to solve for $\delta_m L^H =$ weak form of Hessian action.

In finite dimensions, $\delta_m L^H \Rightarrow \tilde{m}^T H(m) \tilde{m}$ for all $\tilde{m}$

Strong Forms

1) Forward problem:

$$\int_{\Omega} [m \nabla u \cdot \nabla \tilde{p} + \tilde{p} v \cdot \nabla u + \tilde{p} a u^3 - \tilde{p} f] dx =$$

$$= \int_{\Omega} \tilde{p} [-\nabla \cdot (m \nabla u) + v \cdot \nabla u + a u^3 - f] dx + \int_{\Gamma} \tilde{p} m \nabla u \cdot \vec{n} ds = 0 \quad \forall \tilde{p} \in H_0^1$$

arbitrary in Ω

strong form:

solve for
 u
given m

$$\boxed{\begin{aligned} -\nabla \cdot (m \nabla u) + v \cdot \nabla u + a u^3 - f &= 0 \text{ on } \Omega \\ u &= 0 \text{ on } \Gamma \end{aligned}}$$

2) Adjoint problem:

$$\int_{\Omega} [(B u - d) B \hat{u} + m \nabla \hat{u} \cdot \nabla p + p v \cdot \nabla \hat{u} + 3 p a u^2 \hat{u}] dx$$

$$= \int_{\Omega} \hat{u} [B^*(B u - d) + -\nabla \cdot (m \nabla p) + -\nabla \cdot (p v) + 3 p a u^2] dx$$

$$+ \int_{\Gamma} \hat{u} (m \nabla p \cdot \vec{n} + p v \cdot \vec{n}) ds \quad \forall \hat{u} \in H_0^1$$

solve for
 p

$$\boxed{\begin{aligned} -\nabla \cdot (m \nabla p) - \nabla \cdot (p v) + 3 p a u^2 &= -B^*(B u - d) \text{ on } \Omega \\ p &= 0 \text{ on } \Gamma \end{aligned}}$$

given m, u

3) Gradient:

$$\begin{aligned} & \int_{\Omega} [\beta \nabla m \cdot \nabla \hat{m} + \hat{m} \nabla u \cdot \nabla p] dx \\ &= \int_{\Omega} \hat{m} (-\beta \Delta m + \nabla u \cdot \nabla p) dx \quad \forall \hat{m} \in H^1 \\ &+ \int_{\Gamma} \hat{m} \beta \nabla m \cdot \vec{n} ds = (G(m), \hat{m}) \quad "g(m)^T \hat{m}" \end{aligned}$$

given
 m, u, p

$-\beta \Delta m + \nabla u \cdot \nabla p$	on Ω
$\beta \nabla m \cdot \vec{n}$	on Γ

$G(m)$: strong gradient
(m isn't changing on boundary)

4) Steepest Descent Algo in Inf-Dim:

$$(\delta_m, \hat{m}) = -(G(m), \hat{m}) \quad \forall \hat{m} \in H^1$$

steepest descent

search direction, previously p .

$$" \delta_m^T \hat{m} " = " g(m)^T \hat{m} " \quad \forall \hat{m} \in H^1 \quad \square$$

4) Incremental forward problem:

$$\begin{aligned} & \int_{\Omega} [\tilde{m} \nabla u \cdot \nabla \hat{p} + m \nabla u \cdot \nabla \hat{p} + \hat{p} v \cdot \nabla \tilde{u} + 3\hat{p} a u^2 \tilde{u}] dx \\ &= \int_{\Omega} \hat{p} [-\nabla \cdot (\tilde{m} \nabla u) - \nabla \cdot (m \nabla \tilde{u}) + v \cdot \nabla \tilde{u} + 3a u^2 \tilde{u}] dx \\ &+ \int_{\Gamma_0} \hat{p} (\tilde{m} \nabla u \cdot \vec{n} + m \nabla \tilde{u} \cdot \vec{n}) = 0 \quad \forall \hat{p} \in H^1_0 \end{aligned}$$

solve for

$\tilde{u}$,

given $m, \tilde{m}, u$

$-\nabla \cdot (m \nabla \tilde{u}) + v \cdot \nabla \tilde{u} + 3a u^2 \tilde{u} = \nabla \cdot (\tilde{m} \nabla u)$	in Ω
$\tilde{u} = 0$	on Γ

note: linear in $\tilde{u}$; linearized around current u (1)
w/ coefficient $3au^2$

$\tilde{m}$ is known $\Rightarrow$ prescribed by Cb.

Rest of the forward & incremental forward problem (diffusion, advection) is same.

5) Incremental Adjoint problem:

$$\int_{\Omega} [\tilde{m} \nabla \hat{u} \cdot \nabla p + m \nabla \hat{u} \cdot \nabla \tilde{p} + \tilde{p} v \cdot \nabla \hat{u} + 3\tilde{p} a u^2 \hat{u} + B^* \hat{u} B \tilde{u} + 6p a u \hat{u} \tilde{u}] dx = 0 \quad \forall \hat{u} \in H_0^1$$

$\rightarrow$ arbitrary

$$\int_{\Omega} \hat{u} [-\nabla \cdot (\tilde{m} \cdot \nabla p) - \nabla \cdot (m \nabla \tilde{p}) - \nabla \cdot (\tilde{p} v) + 3\tilde{p} a u^2 + B^* B \tilde{u} + 6p a u \tilde{u}] dx$$

$$+ \int_{\Gamma} \hat{u} [\tilde{m} \nabla p \cdot \vec{n} + m \nabla \tilde{p} \cdot \vec{n} + \tilde{p} v \cdot \vec{n}] ds = 0 \quad \forall \hat{u} \in H_0^1$$

solve for $\tilde{p}$, given $m, u, \tilde{m}, \tilde{u}, p$

$$\boxed{\begin{aligned} -\nabla \cdot (m \nabla \tilde{p}) - \nabla \cdot (\tilde{p} v) + 3\tilde{p} a u^2 &= \nabla \cdot (\tilde{m} \cdot \nabla p) - B^* B \tilde{u} - 6p a u \tilde{u} \\ \tilde{p} &= 0 \text{ on } \Gamma \end{aligned}} \quad \text{on } \Omega$$

if incompressible $-\nabla \cdot \tilde{p} v = -\nabla \tilde{p} \cdot v - \tilde{p} \nabla \cdot v$

like in advection $\nabla \cdot v = 0$

similar to adjoint eqn

6) Hessian action $H(m) \tilde{m} : \quad \forall \tilde{m} \in H^1$

$$\int_{\Omega} [\beta \nabla \tilde{m} \cdot \nabla \tilde{m} + \tilde{m} \nabla u \cdot \nabla \tilde{p} + \tilde{m} \nabla \tilde{u} \cdot \nabla p] dx \quad \text{not necessarily 0.}$$

$$= \int_{\Omega} \underbrace{\tilde{m} [-\beta \Delta \tilde{m} + \nabla u \cdot \nabla \tilde{p} + \nabla \tilde{u} \cdot \nabla p]}_{H\tilde{m} \text{ in } \Omega} dx + \int_{\Gamma} \underbrace{\tilde{m} (\beta \nabla \tilde{m} \cdot \vec{n})}_{H\tilde{m} \text{ on } \Gamma} ds$$

$$\boxed{\begin{aligned} -\beta \Delta \tilde{m} + \nabla u \cdot \nabla \tilde{p} + \nabla \tilde{u} \cdot \nabla p &\text{ on } \Omega \\ \beta \nabla \tilde{m} \cdot \vec{n} &\text{ on } \Gamma \end{aligned}} \quad \begin{aligned} &\text{Evaluate } H(m) \text{ on } \tilde{m} \text{ (at } m) \\ &\text{given } m, u, p, \tilde{u}, \tilde{p} \end{aligned}$$

Newton Algo in Inf-Dim

